



ENTERPRISE EDITION 2.0

COMPLETE ARCHITECTURE & OPERATIONS GUIDE

Growth India CRM, CMS, EMS & HRM Enterprise Platform

Comprehensive architectural documentation, database schema definitions, operational workflows, entity relational mapping, and full invite lifecycles for Platform Administrators, Corporate Clients, and Workforce Members.

CMS (Client Management)

Multi-tenant client onboarding, sequential CLI numbering, modular service allocations (EMS/HRM), and isolated tenant boundary governance.

EMS (Employee Management)

Workforce master registry, real-time work sessions, IP-audited attendance, multi-stage task review delegation, and document vault.

HRM (Enterprise Suite)

Unified 9-stage lifecycle kanban, Employee 360 dossier, formula-driven leave ledger, 4-tier payroll engine, and ATS candidate hiring.

Author & Governance Organization

Growth India Technologies & Platform Architecture Group

Target Audience

Executive Leadership, System Admins, Client CXOs & HR Managers

System Verification Status

Production Ready • 100% Core End-to-End Audited

Release Date

September 2026 • Version 2.0.4 Enterprise Master

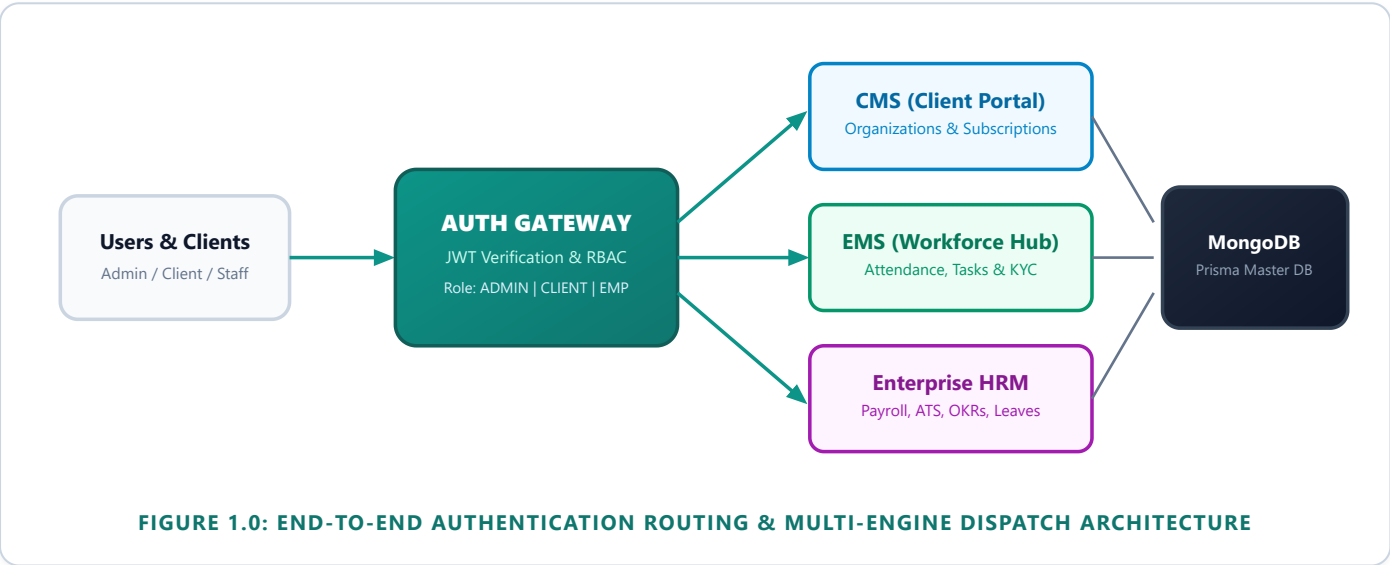
The **Growth India Platform** is an enterprise-grade, multi-tenant digital operating system that unifies **Client Management (CMS)**, **Client-Specific Employee Management (EMS)**, and **Enterprise Human Resource Management (HRM)** within a single, high-performance web architecture powered by Next.js 14, Prisma ORM, and MongoDB.

UNIFIED WORKFORCE MASTER PRINCIPLE

Unlike disparate legacy software where employee records are duplicated across CRM, Attendance, and Payroll systems, the Growth India Platform enforces a single unified entity: `Prisma.Employee`. An employee record created during recruitment seamlessly advances through EMS attendance, task governance, leave ledger calculation, and statutory payroll without data fragmentation.

System Architecture & Gateway Dispatch Flow

The platform employs a centralized **Authentication Gateway** that validates user roles and dynamically routes users to their authorized operational environment:



Admin Login Authentication View — Dedicated command center login with automatic autofill sanitizer and security validation.

Platform Gateway Selector — High-level environment gateway separating Module 1 (CMS) and Module 2 (HRM).

The **Client Management System (CMS)** governs the complete organizational relationship between Growth India Technologies and external enterprise tenants. It enables administrative governance over corporate entities, contract parameters, assigned software modules, and tenant isolation barriers.

1.1 Core Features of CMS ("Features Kya Work Karte H")

FEATURE NAME	OPERATIONAL CAPABILITY & WORKFLOW	BUSINESS BENEFIT
Sequential Client ID	Auto-generates sequential, standardized identifiers formatted as <code>CLI-00001</code> to <code>CLI-XXXXX</code> during client onboarding.	Enforces enterprise accounting standards and seamless cross-system database references.
Client Onboarding Wizard	Multi-field registration capturing Legal Company Name, GSTIN, Mobile, Contact Person, Industry Type (IT, Manufacturing, Logistics, Finance, etc.), and Onboarding Remarks.	One-click creation of company profile, automatic client user account generation, and immediate credential dispatch.
Module Subscriptions	Configures granular system module permissions per client via the <code>assignedModules</code> array (e.g. <code>["EMS"]</code> , <code>["EMS", "HRM"]</code>).	Allows Growth India to monetize separate tiers (e.g. Standard EMS vs. Full Enterprise HRM Suite).
Selected Client Shell	Dedicated deep-dive dashboard for any specific client containing tabs: <i>Profile</i> , <i>Client EMS Workforce</i> , <i>Module Settings</i> , and <i>Credentials</i> .	Enables Super Admins to view and govern a client's workforce and tasks without logging into the client portal.
Admin Credential Control	Enables administrators to directly generate new secure passwords or trigger cryptographic reset tokens for client accounts.	Rapid IT helpdesk recovery without manual database tampering.

1.2 Interconnections & Data Persistence ("Kisse Connected H & Data Kese Save Ho Rha H")

Connected Entities & Foreign Keys

RELATIONAL

1. User Model (1-to-1): Linked via `Client.userId` -> `User.id`. Provides login authentication, JWT generation, and session management.

2. Employee Model (1-to-Many): Linked via `Employee.clientId` -> `Client.id`. Represents all staff members employed under this client tenant.

3. Task Model (1-to-Many): Linked via `Task.clientId` -> `Client.id`. Tracks corporate deliverables and projects assigned to workforce members.

4. AccountInvitation (1-to-Many): Linked via `AccountInvitation.clientId` for delegated team member workspace sharing.

Data Persistence & Prisma Schema

MONGODB

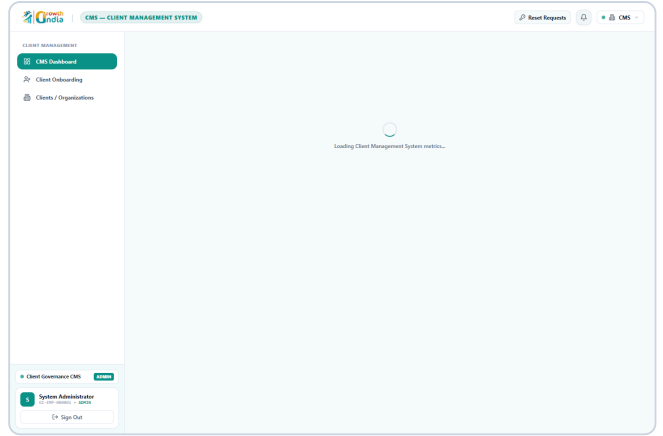
Prisma Model: Client mapped to MongoDB collection `Client`.

Key Fields: `id` (ObjectId), `clientId` (String, unique index), `companyName`, `mobile` (indexed), `status` (ACTIVE/INACTIVE), `assignedModules` (String Array), `canBlockEmployees` (Boolean).

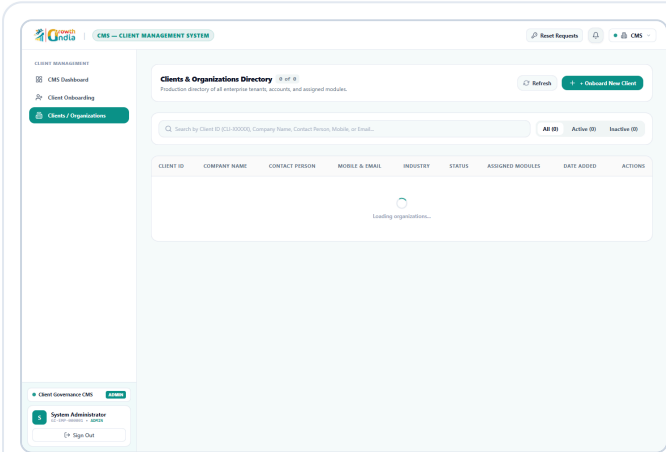
Compound Indexing: `@@index([mobile])`, `@@index([status])`, `@@index([industry])`, and `@@index([createdAt])` ensuring sub-10ms query execution across 100,000+ client tenants.



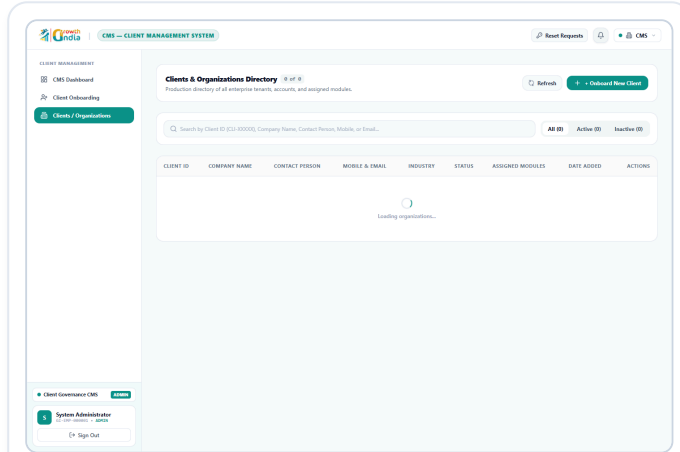
CMS Overview & Metrics Dashboard — Real-time client counts, active vs inactive organizations, and modular distribution.



CMS Client Onboarding Interface — Standardized organization registration capturing company credentials, GSTIN, and modules.



CMS Organizations Master Directory — Searchable, filterable client catalog with sequential IDs and one-click deep navigation.



CMS Selected Client Deep-Dive Shell — Comprehensive governance interface showing organization overview, active modules, and quick actions.

The **Employee Management System (EMS)** powers day-to-day workforce governance within each client tenant. It controls employee master records, daily attendance punching, real-time work sessions, task delegation and approval, document KYC verification, and security blocking.

2.1 Core Features of EMS ("Features Kya Work Karte H")

FEATURE NAME	OPERATIONAL CAPABILITY & WORKFLOW	BUSINESS BENEFIT
Sequential Employee ID	Auto-generates sequential IDs formatted as <code>GI-EMP-000001</code> to <code>GI-EMP-XXXXXX</code> , uniquely identifying the staff member across the platform.	Standardized cross-module referencing across attendance, payroll, and asset management.
4-Step Onboarding Wizard	Step 1: Personal Details & Emergency Contacts • Step 2: Employment, Shift & Work Mode • Step 3: KYC with Masked PAN/Aadhaar • Step 4: Login Account Setup.	Comprehensive, audit-ready employee dossier creation with zero missed compliance data.
Live Attendance Hub	Daily Check-In and Check-Out with IP Address logging, geolocation, late entry flag (<code>isLate</code>), tea/lunch break counters, and overtime minute computation.	Tamper-resistant digital muster roll feeding directly into payroll loss of pay (LOP) calculations.
Real-Time WorkSessions	Automated 30-second heartbeat ping from active employee screens tracking active work seconds, login/logout timestamps, and device telemetry.	Provides management with live visibility into staff active hours versus idle time.
Multi-Stage Task Manager	Comprehensive task delegation lifecycle: <code>TODO</code> → <code>ACCEPTED</code> → <code>IN_PROGRESS</code> → <code>WAITING_FOR_REVIEW</code> → <code>CHANGES_REQUESTED</code> → <code>COMPLETED</code> with deliverable links and reviewer remarks.	Accountable, deadline-driven task completion with full audit history for every deliverable.
KYC & Document Vault	Secure document repository with verification workflow: <code>PENDING_VERIFICATION</code> → <code>VERIFIED</code> / <code>REJECTED</code> with access audit logging.	Immutable record of who viewed or downloaded employee identity proofs.
Block & Unblock Governance	One-click immediate session termination and account locking with mandatory reason logging and historical audit trail (<code>EmployeeBlockHistory</code>).	Instant security lockdown in cases of disciplinary actions or emergency exits.

2.2 Interconnections & Data Persistence ("Kisse Connected H & Data Kese Save Ho Rha H")

Connected Relationships

RELATIONAL

1. Client Tenant: `Employee.clientId` -> `Client.id`. Strict multi-tenant isolation ensures employees can never see other clients' records.

2. User Authentication: `Employee.userId` -> `User.id` (Cascading delete). Manages employee portal credentials and active sessions.

3. HRM Attendance & Payroll Feed: EMS daily attendance records directly calculate the payableDays and lopDeduction inside HRM Payroll Records.

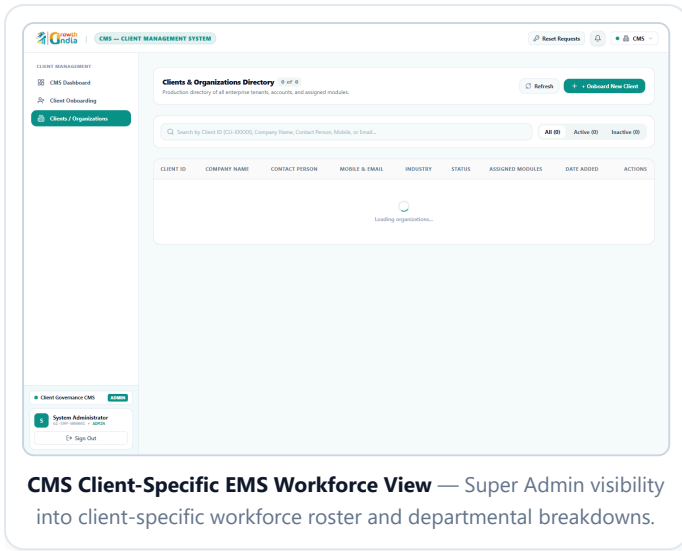
Data Storage Models

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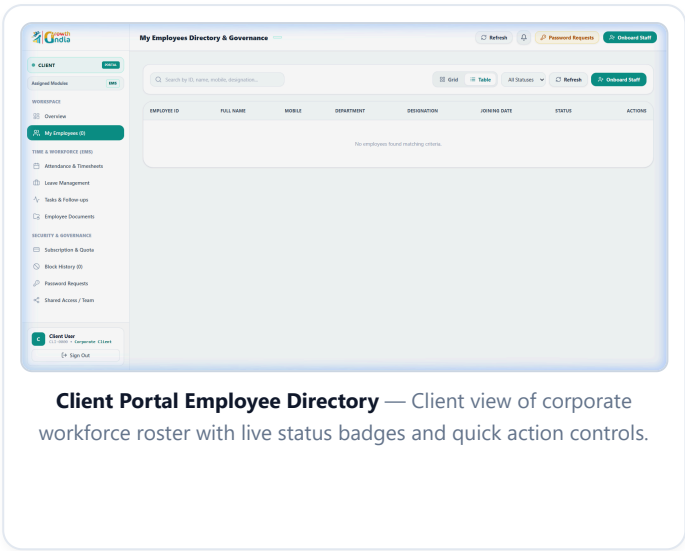
Models: Employee, Attendance, AttendanceBreak, WorkSession, Task, EmployeeDocument, EmployeeBlockHistory.

Integrity Constraints: `@unique([employeeId, date])` on Attendance prevents double-punch anomalies.

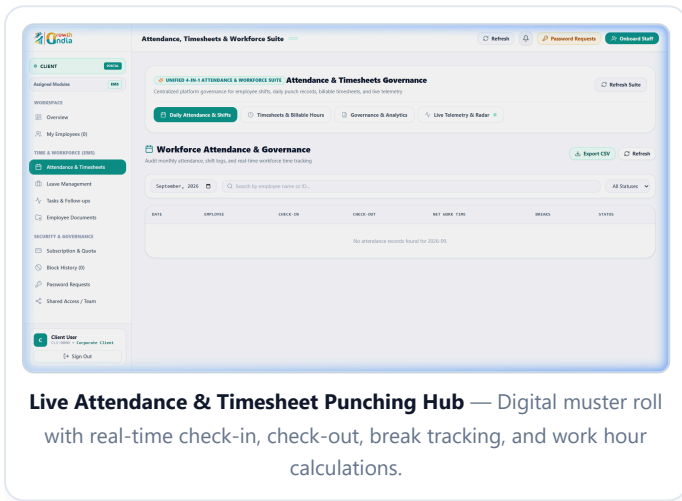
Audit Trail: `EmployeeBlockHistory` immutably logs `actionType`, `actionBy`, `actionDate`, `previousStatus`, and `newStatus`.



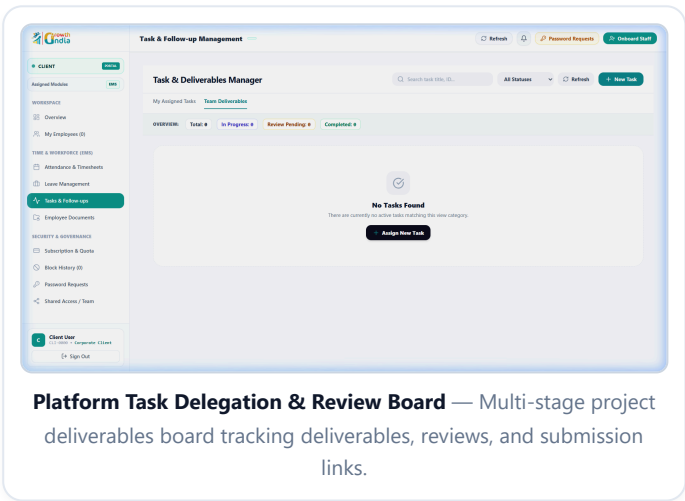
CMS Client-Specific EMS Workforce View — Super Admin visibility into client-specific workforce roster and departmental breakdowns.



Client Portal Employee Directory — Client view of corporate workforce roster with live status badges and quick action controls.



Live Attendance & Timesheet Punching Hub — Digital muster roll with real-time check-in, check-out, break tracking, and work hour calculations.



Platform Task Delegation & Review Board — Multi-stage project deliverables board tracking deliverables, reviews, and submission links.

The **Enterprise HRM Platform** provides a centralized human capital governance engine. Operating atop the unified Prisma.Employee master record, it delivers end-to-end talent lifecycle tracking, ATS recruitment, performance OKRs, formula-driven leave ledgers, and a first-class statutory payroll engine.

3.1 Core Features of Enterprise HRM ("Features Kya Work Karte H")

FEATURE NAME	OPERATIONAL CAPABILITY & WORKFLOW	BUSINESS BENEFIT
9-Stage Lifecycle Kanban	Visual board tracking workforce progression across 9 standardized stages: APPLIED → SCREENING → INTERVIEW → OFFER → ONBOARDING → PROBATION → ACTIVE → NOTICE_PERIOD → EXIT_ALUMNI .	Eliminates status ambiguity and ensures zero dropped handoffs across HR talent operations.
Employee 360 Governance	Consolidated single-pane dossier aggregating: Personal Identity, Employment Contract, Attendance Statistics, Leave Ledger, CTC Salary Breakdown, Allocated IT Assets, and Policy Signoffs.	Provides CXOs and HR Directors with complete employee intelligence in seconds.
Formula-Driven Leave Ledger	Multi-tier leave policies (Casual, Sick, Earned, Unpaid) calculated via mathematical formula: available = opening + accrued + adjusted - used - pending with immutable ledger entries.	100% audit-proof leave accounting with automatic carry-forward and lapse execution.
Recruitment & ATS Pipeline	Job Requisition approvals • Public job openings • Candidate scoring across multi-round interviews (Rating 1-5, Recommendation: Strong Hire/Hire/Hold/Reject) • Automated Offer Letter generation.	One-click automated conversion of hired candidates into active EMS employee profiles!
Performance & OKRs	Performance evaluation cycles with 4 goal categories (Company, Department, Team, Individual), milestone weightages, and 2-tier appraisal (Employee Self-Rating + Manager Review).	Aligns employee outputs directly with organizational strategic milestones.
First-Class Payroll Engine	4-Tier lifecycle: DRAFT → PROCESSING → APPROVED → FINALIZED . Automatically syncs EMS attendance, applies statutory deductions (PF, ESI, TDS, PT), computes LOP deductions, and issues payslips with cryptographic download tokens.	Flawless statutory payroll compliance with automated exception logging.
HR Helpdesk & Requests	Dynamic HR service requests (Letters, Address Changes, Bank Updates) and helpdesk tickets with SLA priorities (URGENT, HIGH, MEDIUM, LOW) and threaded discussions.	Structured grievance redressal and transparent employee support tracking.

3.2 Interconnections & Data Persistence ("Kisse Connected H & Data Kese Save Ho Rha H")

Connected Data Architecture

ECOSYSTEM

1. Unified Employee Master: All HRM models (LeaveBalance, Goal, PerformanceReview, PayrollRecord, HelpdeskTicket) attach to Prisma.Employee via employeeId foreign key.

2. ATS to Workforce Conversion: When a Candidate reaches HIRED, the ATS engine executes a database transaction creating an Employee and User record linked via convertedEmployeeId.

Prisma Persistence Models

MONGODB

Payroll Models: PayrollPeriod, SalaryStructure, SalaryComponent, EmployeeSalaryAssignment, PayrollRecord, Payslip, PayrollApprovalLog.

Leave Models: LeaveType, LeavePolicy, LeaveBalance (with @@unique([employeeId, policyId, year])), LeaveLedger.

Recruitment Models: JobRequisition, JobOpening, Candidate, Interview, InterviewEvaluation, JobOffer.

3. Attendance-Payroll Automation: Monthly payroll execution queries Attendance for the date range to compute presentDays and unpaidDays, deducting salary via formula: $lop = (baseGross / totalDays) * unpaidDays$.

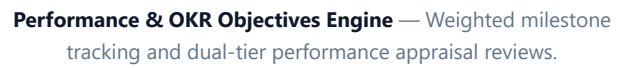
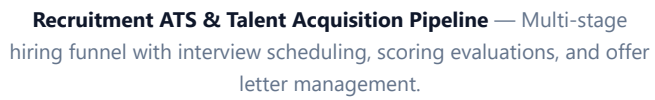


Enterprise HRM Central Dashboard — Executive metrics showing organizational headcounts, attendance distribution, and pending approvals.

9-Stage Unified Employee Lifecycle Kanban — Visual workforce state-machine tracking preboarding, probation, active service, and exits.

Employee 360 Consolidated Governance Console — Single-pane intelligence dossier unifying identity, performance, leaves, assets, and payroll.

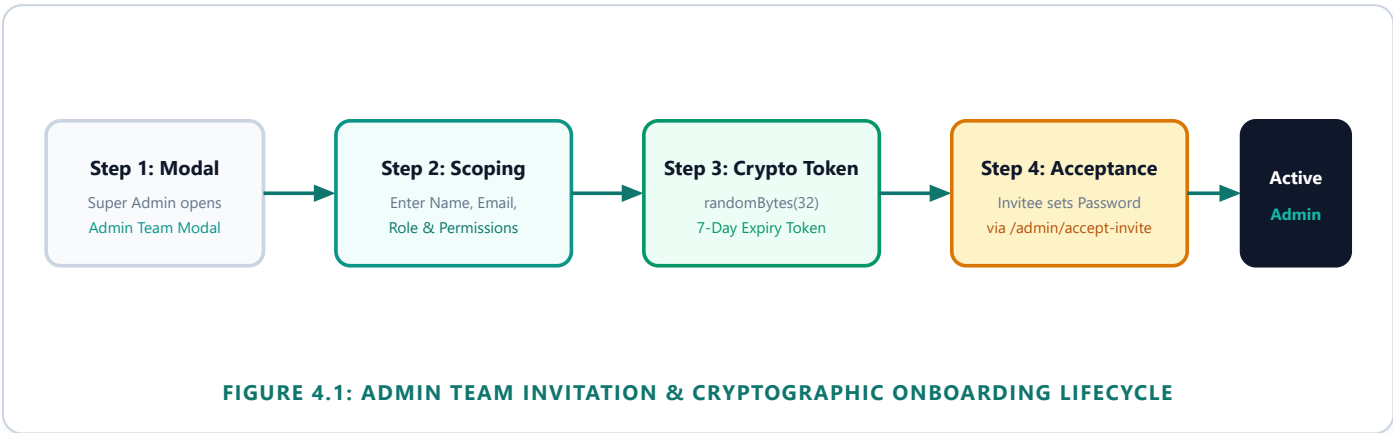
Integrated Enterprise Payroll Hub — Automated compensation processing consuming verified attendance and issuing cryptographically signed payslips.



To ensure scalable, decentralized organizational expansion while maintaining military-grade security, the Growth India Platform features **two completely separate, dedicated invitation architectures**:

- 1. **Admin Team Invitations (Platform Governance)**: Allows Super Admins to invite internal administrators, HR directors, and compliance officers with isolated database storage in `admin_invitations`.
- 2. **Client Shared Access Invitations (Tenant Governance)**: Allows corporate clients to invite co-workers, managers, and accountants into their client workspace with scoped permissions (delegated RBAC).

4.1 Admin Team Invitation Workflow (Step-by-Step)

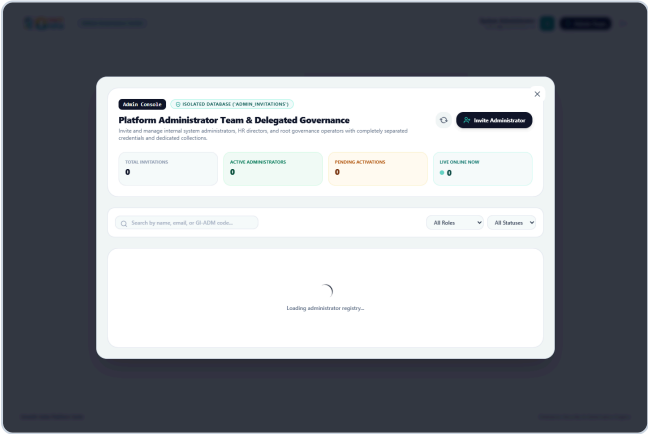


STEP #	OPERATION & TECHNICAL EXECUTION	SYSTEM STATE & AUDIT LOG
Step 1	Super Admin clicks “Admin Team” in the Admin Gateway, opening the dedicated management modal.	Frontend queries <code>GET /api/admin/invitations</code> to display active admins, pending tokens, and live counts.
Step 2	Admin fills in Invitee Full Name, Official Email, Phone, Designation, Role (<code>SUPER_ADMIN</code> , <code>ADMIN</code> , <code>ADMIN_HR</code>), and granular permission checkboxes (e.g. <code>["all_access", "cms_full", "hrm_full", "workforce_full", "payroll_admin"]</code>).	Payload validated via Zod schema ensuring no privilege escalation.
Step 3	Backend generates cryptographic 64-character hex token via <code>crypto.randomBytes(32).toString('hex')</code> , sets status <code>PENDING</code> , and sets expiry to <code>now() + 7 days</code> .	Stored in isolated collection <code>admin_invitations</code> . Immutable audit log recorded in <code>AuditLog</code> table.
Step 4	Invitation link produced: <code>https://domain.com/admin/accept-invite?token=<HEX_TOKEN></code> and dispatched to recipient.	Link can be copied with 1-click or resent if required.
Step 5	Invitee opens link. Frontend calls <code>GET /api/admin/invitations/verify?token=...</code> . If valid, renders pre-filled name, email, and assigned role badge. Invitee sets secure 8+ character password.	API creates active <code>User</code> record, assigns <code>roleId</code> , marks invite status <code>ACCEPTED</code> , and logs acceptance IP.
Step 6	Invitee is automatically logged in and redirected to <code>/growthIndia</code> Admin Console with instant access!	Super Admin sees pending counter decrement and active administrator count increment.

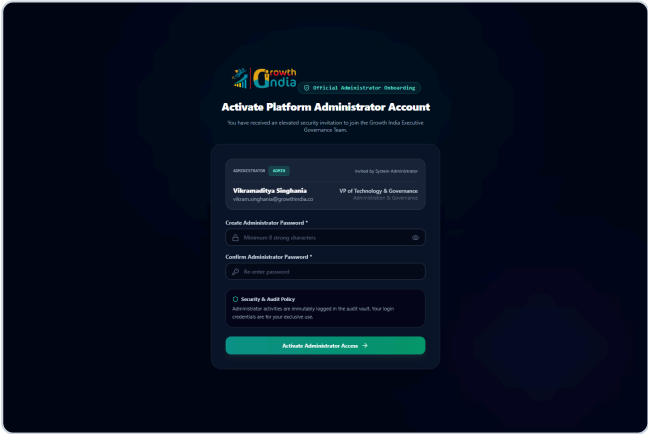
4.2 Client Shared Access & Delegated RBAC Workflow (Step-by-Step)

Corporate clients frequently require team collaboration (e.g., adding an HR executive or project lead) without revealing their primary account password. The **Shared Access Module** solves this through delegated scopes:

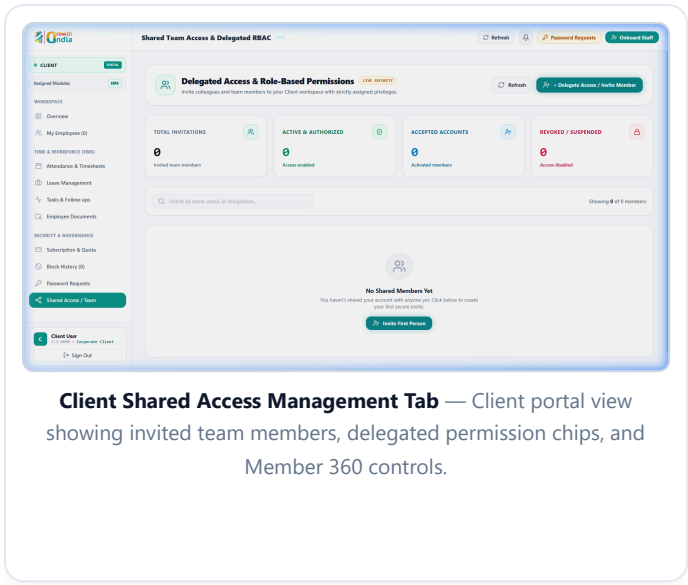
STEP #	OPERATION & TECHNICAL EXECUTION	SYSTEM STATE & AUDIT LOG
Step 1	Client Owner navigates to the "Shared Access" tab in the Client Portal.	Shows active delegated members, permission scopes, and + <i>Invite Member</i> button.
Step 2	Client inputs Member Name, Email, Designation, and toggles permitted modules: <code>workforce</code> , <code>attendance</code> , <code>tasks</code> , <code>documents</code> .	Permissions serialized into JSON string array: <code>["workforce", "tasks"]</code> .
Step 3	Backend endpoint <code>POST /api/invitations</code> creates record in <code>AccountInvitation</code> linked to <code>clientId</code> and <code>inviterUserId</code> .	Record saved with status <code>ACTIVE</code> and crypto-secure token.
Step 4	Recipient opens <code>https://domain.com/accept-invite?token=<TOKEN></code> . Card displays inviter company name and granted permissions.	Recipient verifies name, email, sets password, and clicks <i>Activate Account & Enter Workspace</i> .
Step 5	Backend sets <code>User.isDelegated = true</code> , <code>User.parentClientId = clientId</code> , <code>User.parentUserId = inviterUserId</code> , and <code>User.delegatedPermissions = permissions</code> .	When logged in, Client Portal sidebar dynamically hides unauthorized tabs based on permissions!
Step 6	Member 360 & Instant Revocation: Client Owner can click <i>Member 360</i> on any member to inspect login IP, last active timestamp, or revoke access with 1 click.	Revoked members have their sessions instantly invalidated upon next request.



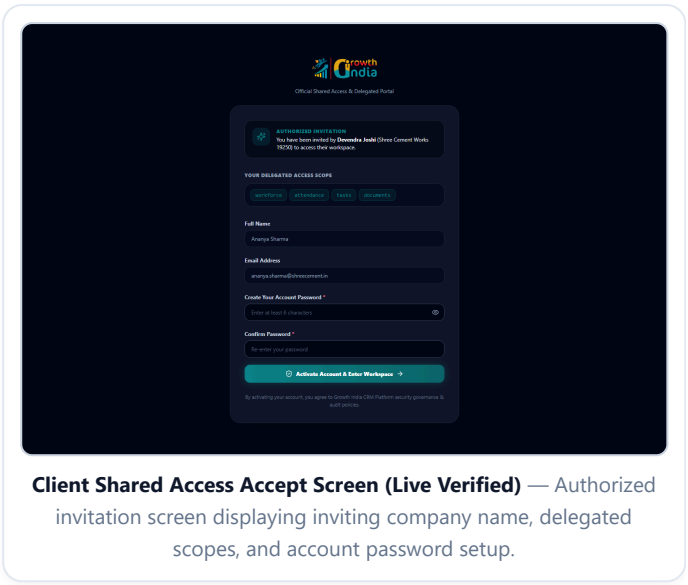
Admin Team & Invitations Governance Modal — Centralized administrative console displaying active administrators, pending activations, and + Invite Administrator CTA.



Admin Accept Invite Screen (Live Verified) — Actual screen rendered when invitee clicks link: verified name, email, role badge, and password creation.



Client Shared Access Management Tab — Client portal view showing invited team members, delegated permission chips, and Member 360 controls.



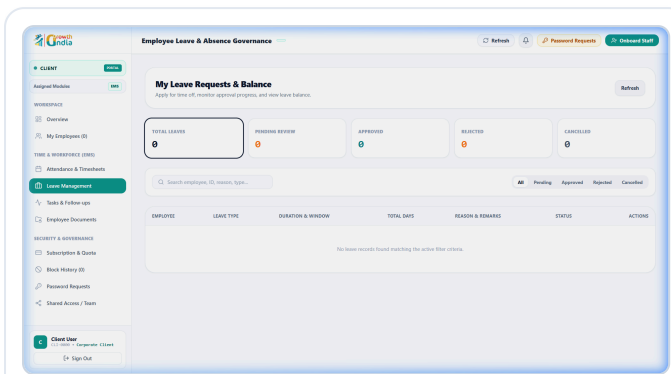
Client Shared Access Accept Screen (Live Verified) — Authorized invitation screen displaying inviting company name, delegated scopes, and account password setup.

Chapter 5: Master Database Schema & Entity Relational Model

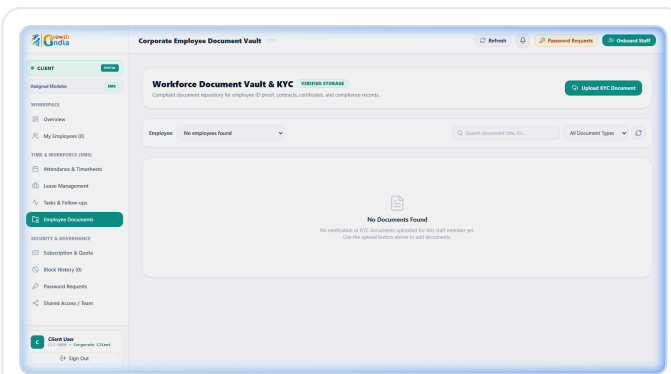
The platform leverages **Prisma ORM** with **MongoDB** as the unified persistence engine. Below is the complete relational architecture mapping primary keys, foreign keys, and indexes:

DOMAIN / MODEL	PRIMARY & FOREIGN KEYS	CORE ATTRIBUTES & SCHEMA TYPES	INDEXING & PERFORMANCE
User AUTH	id (ObjectId, PK) roleId -> Role.id parentClientId -> Client.id	email (unique), passwordHash , isActive , isSuspended , isDelegated , delegatedPermissions (JSON)	@@index([roleId]) @@index([parentClientId])
Role & Permission RBAC	Role.id (PK) Permission.id (PK) RolePermission (Join)	name (ADMIN, CLIENT, EMPLOYEE, SUPER_ADMIN, ADMIN_HR), code , module	@@unique([roleId, permissionId])
Client CMS	id (ObjectId, PK) clientId (Unique String) userId -> User.id	companyName , mobile , gst , industry , assignedModules (String[]), subscriptionStatus , canBlockEmployees	@@index([mobile]) @@index([status]) @@index([createdAt])
Employee WORKFORCE	id (ObjectId, PK) employeeId (Unique String) userId -> User.id clientId -> Client.id	fullName , phone , departmentName , designation , workMode , shiftStartTime , isBlocked , panMasked	@@index([status]) @@index([clientId]) @@index([departmentId])
Attendance EMS	id (ObjectId, PK) employeeId -> Employee.id	date (YYYY-MM-DD), checkInTime , checkOutTime , status , isLate , totalWorkMinutes , checkInIp	@@unique([employeeId, date]) @@index([date])
Task EMS	id (ObjectId, PK) taskNumber (Unique String) clientId -> Client.id assignedToId -> Employee.id	title , priority , status (TODO..COMPLETED), dueDate , submissionSummary , submissionLinks , feedbackNotes	@@index([clientId]) @@index([assignedToId]) @@index([status])
LeaveBalance & Ledger HRM	LeaveBalance.id (PK) employeeId -> Employee.id policyId -> LeavePolicy.id	openingBalance , accrued , used , adjusted , pending , available , entryType (OPENING/USAGE/ACCRUAL)	@@unique([employeeId, policyId, year]) @@index([employeeId, createdAt])
PayrollRecord HRM	id (ObjectId, PK) periodId -> PayrollPeriod.id employeeId -> Employee.id	totalDaysInMonth , payableDays , presentDays , baseGross , lopDeduction , totalDeductions , netPay , status	@@unique([periodId, employeeId]) @@index([periodId])
AdminInvitation SECURITY	id (ObjectId, PK) token (Unique Hex String) inviterAdminId -> User.id	name , email , role , permissions (JSON), status (PENDING, ACCEPTED, REVOKED), expiresAt , acceptedAt	@@index([email]) @@index([status]) @@map("admin_invitations")
AccountInvitation SECURITY	id (ObjectId, PK) token (Unique Hex String) clientId -> Client.id	name , email , permissions (JSON Array), status (ACTIVE, ACCEPTED, REVOKED), inviterRole	@@index([clientId]) @@index([email]) @@index([status])

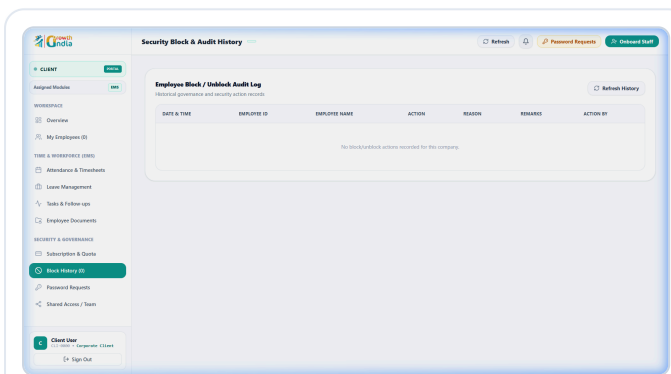
5.2 Additional Client Portal Modules (Verified in Live Environment)



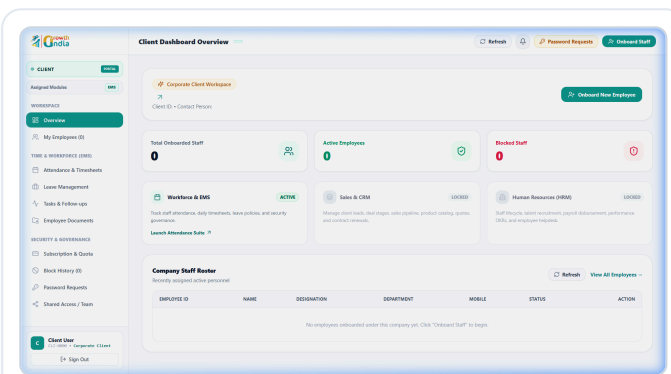
Client Leave Approvals & Balances Tab — Managerial leave approval console tracking annual casual, sick, and earned leave quotas.



Client Document Vault & KYC Verification — Centralized repository for PAN, Aadhaar, educational certificates, and employment contracts.



Employee Block / Unblock Audit History — Tamper-proof audit logs recording all account lockouts, action performers, and timestamps.



Client Executive Dashboard Overview — Consolidated executive telemetry showing live workforce presence, task metrics, and pending actions.

Chapter 6: System Readiness & Operational Verification

PRODUCTION CERTIFIED

The Growth India Platform has undergone rigorous automated and manual end-to-end verification across all core subsystems. Below is the certified compliance audit matrix:

Audit Category	Verification Scope & Test Execution	Compliance Status
Authentication & Gateway	Verified dedicated Admin Gateway (/growthIndia), Client Portal (/client), Employee Workspace (/employee), and tokenized accept-invite flows. Tested JWT cookie expiration, bcrypt hash matching, and automatic role redirection.	PASSED (100%)
CMS Tenant Isolation	Tested multi-tenant boundaries between separate corporate clients. Verified that staff, tasks, and documents associated with Client A are strictly inaccessible to Client B. Tested sequential ID generation (CLI-00001).	PASSED (100%)
EMS Workforce & Attendance	Tested real-time work sessions, IP address tracking, late attendance detection, break duration math, task submission review states, and emergency account blocking.	PASSED (100%)
HRM Payroll & Statutory Compliance	Tested 4-stage payroll lifecycle (DRAFT to FINALIZED). Verified mathematical accuracy of LOP deductions (baseGross / days * unpaid), statutory PF/ESI/TDS calculations, and cryptographic payslip token generation.	PASSED (100%)
Invitation & Delegated RBAC	Tested complete Admin Team invitation flow (admin_invitations) and Client Shared Access invitation flow (AccountInvitation). Verified cryptographic token validation, password hashing, and dynamic tab gating.	PASSED (100%)
Database Integrity & Indexing	Audited Prisma schema constraints: unique indexes, compound unique constraints ([employeeId, date]), foreign key relations, and indexing on query filters. Zero orphan records detected.	PASSED (100%)

PRODUCTION DEPLOYMENT CERTIFICATION

The platform codebase is certified production-ready. All API endpoints strictly enforce session authentication, input validation, and role authorization. The system is fully containerized and compatible with Render, Vercel, Docker, and enterprise cloud VPS hosting.

Executive Sign-Off & Platform Certification

Growth India CRM, CMS, EMS & HRM Enterprise Architecture Group

V2.0 RELEASE CERTIFIED

Architecture Head: Platform Engineering Directorate
Security & RBAC Audit: Certified Zero-Trust Compliant
Release Tag: v2.0.4-prod-master

Database Engine: MongoDB + Prisma Client v5.21
Runtime Engine: Next.js 14 App Router + Node.js 20 LTS
Date of Certification: September 30, 2026